

Interview Protocols for Healthcare-oriented iDEC Inclusivity Research

Semi-structured interview guides for front-line clinical and community stakeholders — designed for biomedical iDEC teams working on diagnostic, therapeutic, or probiotic projects.

LZU-CHINA · GutSentry · Open Community Resource · Interview Protocols

How to use these protocols

These are semi-structured guides, not scripts. Treat them as a checklist of topics you must cover; the exact wording is up to you and your interviewer. The goal is to make the stakeholder feel like they are talking to a colleague, not filling in a form.

- Pick the guide that matches your stakeholder (nurse / lab staff / disability or public-health NGO / patient-family proxy).
- Send 3–5 questions in advance so they can prepare — busy clinicians will thank you.
- Keep it 25–40 minutes. Longer than 45 minutes and you lose them.
- Record only with written consent. If they say no, take handwritten notes and summarise back to them.
- Always end by asking: "What did we not ask that we should have?" — this is where the real gold hides.

NOTE · These guides were developed by LZU-CHINA during the 2026 season for our EcN-based gut-infection diagnostic-therapeutic project (GutSentry). They are shared as-is for other teams to adapt. Replace every [bracketed] term with your own project detail.

Guide A · ICU / Geriatric / Ward Nursing Staff

Use this when talking to nurses, ward aides, or geriatric-care staff. These people see the patient workflow up close — they know where a clever device will silently fail.

Opening (about 2 minutes)

Thanks for making time. We are a student synthetic-biology team, and we are building [one-line project description, e.g. a probiotic that changes colour when it detects a gut infection]. We are not here to sell anything — we just want to know where our idea would fall apart in your day. Anything you say stays anonymous, and we will not use your name in our report.

Core questions (pick 6–8, in any order)

1. Walk us through a typical day when a patient under your care develops a gut infection. Where does the diagnosis usually start, and how long does it take?
2. If our [device / probiotic / kit] were sitting on your ward tomorrow, what is the very first thing that would go wrong? Be honest — we would rather hear it now.
3. Lighting, noise, interruptions — what is the ward actually like when you are trying to read a test result? Can you describe a time when a colour change was hard to read?
4. Who actually collects the sample — you, the patient, a family member? How reliable is that, especially with elderly or immobile patients?
5. What does "simple" mean to you? If we said our kit has three steps, would that feel simple or already too many?
6. What scares patients or families most about a new infection? What do they ask you about?
7. If you could change one thing about the tools you currently use for this, what would it be?
8. What would make you trust — or refuse to use — something made by students, or something based on live bacteria?

Closing (about 2 minutes)

That was really useful. One last thing — is there anyone else we should talk to? And is there a question we should have asked but did not?

NOTE · What we learned (LZU-CHINA): ward lighting is often dim, colour-based readouts need high contrast, and multi-step sample prep is unrealistic for critically ill patients. This pushed us to specify a high-contrast chromogen and a single-step sample workflow.

Guide B · Grassroots / County Hospital Laboratory & Public-health Staff

Use this for lab technicians, lab directors, county-hospital doctors, or grassroots public-health workers. They will tell you what is actually available outside a well-funded teaching hospital.

Opening (about 2 minutes)

We are a student team working on [project]. We know our idea will mostly land in resource-limited settings, so we wanted to hear from people who actually work there. We are not testing you — we want to know what we are getting wrong.

Core questions (pick 6–8)

9. What diagnostic tools do you currently have for [target disease / condition]? What do you wish you had but cannot get?
10. When a sample comes to your lab, what is the bottleneck — cost, time, training, equipment, or something else?
11. How much does a typical test cost your patients out of pocket? Is that a real barrier?
12. What would a "good" low-resource diagnostic look like for you — budget, steps, interpretation, turnaround time?
13. How do you explain a positive or negative result to a patient or a junior colleague? Is a colour the easiest thing to read, or would a sound or a number be better?
14. What kind of training would your staff need to use something new? How long would they actually sit through?
15. What usually happens to a prototype that comes from a university — does it get used, or does it sit on a shelf?
16. If you could send one message to a team like ours, what would it be?

Closing

Thank you. Is there someone at a village clinic or township health centre we should also hear from? And — what is the one thing we should not assume?

NOTE · What we learned (LZU-CHINA): grassroots labs lack mNGS and qPCR and cannot justify bulky equipment; staff asked for workflows with fewer instruments and fewer steps. This confirmed our point-of-care, instrument-free direction.

Guide C · Disability-health & Patient-advocacy NGOs / Community Organisations

Use this for disability-rights groups, patient associations, Red Cross or Red Crescent branches, community health centres, and similar organisations. These are proxy stakeholders — they speak with, not for, the people your project will serve.

Opening (about 2 minutes)

We are students building [project]. We know we probably have not thought hard enough about who gets left out by a design like ours, so we came to you. We want to listen first. Nothing here is promotional — and if we say something clumsy, please correct us.

Core questions (pick 6–8)

17. From your experience, which groups get excluded most often by medical devices or health information? Where do things break — cost, physical access, language, trust, or something else?
18. If our output is just a colour on a card, who cannot use it, and what would an alternative look like — sound, texture, braille, a phone app, something else?
19. How do you usually explain health information to people with low literacy or limited science background? What works and what falls flat?
20. What worries your community most about anything "engineered" or "genetically modified" in medicine? How would you recommend we talk about it?
21. What would make one of our public-science materials actually useful to you — a flyer, an audio clip, a poster, a workshop?
22. If we made our project open and reusable, would your organisation actually use it? What would stop you?
23. Who in your community should we be talking to but probably are not?
24. What does "inclusion" look like in practice for you — beyond just translating a pamphlet?

Closing

We really appreciate this. Before we go — is there a way we could work with you next time, instead of just asking questions? And is there anything we said today that you would like us to take back or rephrase?

NOTE · What we learned (LZU-CHINA): a colour-only readout excludes visually impaired users; NGO partners also asked for braille and affirmed our audio materials. We added tactile and odour-based readout as optional modules in our design specification.

Guide D · Patient-family members & informal caregivers (proxy stakeholders)

Only use this when direct patient interviews are not feasible (clinical ethics, consent capacity, ward constraints). Family and caregivers are proxies — triangulate their input against literature and clinical staff, never treat it as the patient voice itself.

Core questions

25. Can you tell us about a time your family member had a [gut / infection-related] issue — what was the hardest part for you as a caregiver?
26. What did you understand about the diagnosis at the time? What felt clear, and what felt confusing?
27. If a doctor offered a new test based on a probiotic, what would your first question be?
28. How easy was it to collect or handle a stool sample at home? What made it difficult?
29. Would cost be a deciding factor for you? At what price point would you hesitate?
30. If the result came back as a colour, would you feel confident reading it, or would you want someone to confirm it?

NOTE · Ethics reminder (LZU-CHINA): we did not interview critically ill patients directly. We used proxy stakeholders and cross-validated their input against published clinical literature and nurse interviews. If you plan to interview patients directly, obtain ethics-committee approval and use an accessible, easy-read information sheet.

Before-you-go checklist

Print this, tick it, and put it in your folder. It takes five minutes and prevents the three mistakes teams make most often: no consent form, no plain-language sheet, and nobody taking notes.

Done?	Item	Why it matters
☐	Plain-language information sheet, printed (Chinese + local language if needed)	People cannot consent properly to something they do not understand
☐	Short consent form with two tick boxes (take part / be recorded)	Two separate questions — never bundle them
☐	Anonymisation plan: where will names and recordings be stored, and when deleted	Required by most university data-protection rules
☐	Three questions sent to the participant in advance	Busy clinicians prepare better and talk more freely
☐	Two team members available: one asks, one takes notes	You will miss half of what is said otherwise
☐	Recording device tested, with spare batteries / charged cable	A failed recording wastes everyone's time
☐	A quiet room or a clear online backup plan	Wards and offices are noisy; have a plan B
☐	A one-page debrief template ready to send within 48 hours	Closes the loop and catches mistakes early

Interview logistics & consent (for the team running the session)

The checklist above covers the practical side. The principles below are the ones judges and ethics reviewers will look for — write them into your own process, do not just nod at them.

- Recruit through your university supervisor, a hospital department, or an organisation — never cold-call patients.
- Prepare an information sheet in plain language (Chinese and local language if needed) and a short consent form.
- Ask two questions up front: "Do you agree to this being recorded?" and "Are you comfortable with us quoting you anonymously?"
- Two team members per interview: one asks, one takes notes. Switch roles halfway.

- Within 48 hours, write a one-page summary and send it back for correction if possible — this catches misunderstandings early.
- Store recordings and consent forms separately from the transcript, on a password-protected drive. Delete raw audio after final report.

That is it. Run four or five interviews across Guides A–D and you will have more than enough to shape a credible inclusivity section. Do not aim for volume — aim for contradiction. If two stakeholders disagree, that is the most useful thing you will find.